JOHANNES KEPLER
UNIVERSITY LINZ

General Information



Simon Dorrer

KV VLSI Design, 336.051, WiSe 26/27

INSTITUTE FOR
INTEGRATED CIRCUITS &
QUANTUM COMPUTING

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4040 Linz, Österreich
jku.at

Introduction



About Me – Simon Dorrer

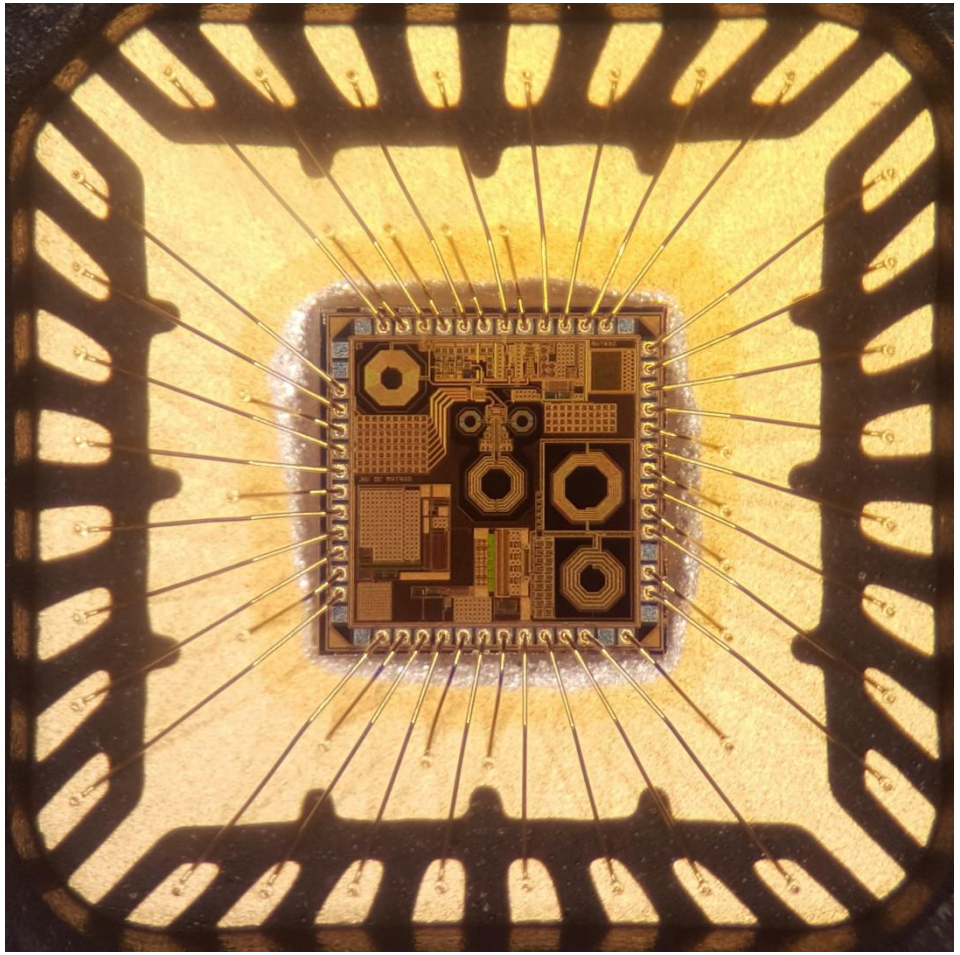
- PhD student @ JKU
- Research and teaching assistant @ IICQC
- Open-source projects:
  - Agentic AI AMS and RF design flows
  - mm-wave six-port receivers
  - TinyWhisper SoC
  - Event-based ADCs
  - Tutorials and templates
- Email: simon.dorrer@jku.at



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Institute for Integrated Circuits and Quantum Computing

- Departments:
  - Integrated Circuits (Prof. Pretl)
  - Quantum Computing (Prof. Kueng)
- Design of analog, digital, and RF integrated circuits (e.g. for smartphones, Internet of Things, 6G, medical electronics, ADCs, ...)
- Design, verification, and testing of circuits and systems
- Open-source tool development for chip design https://github.com/iic-jku/iic-osic-tools
- Agentic AI chip design

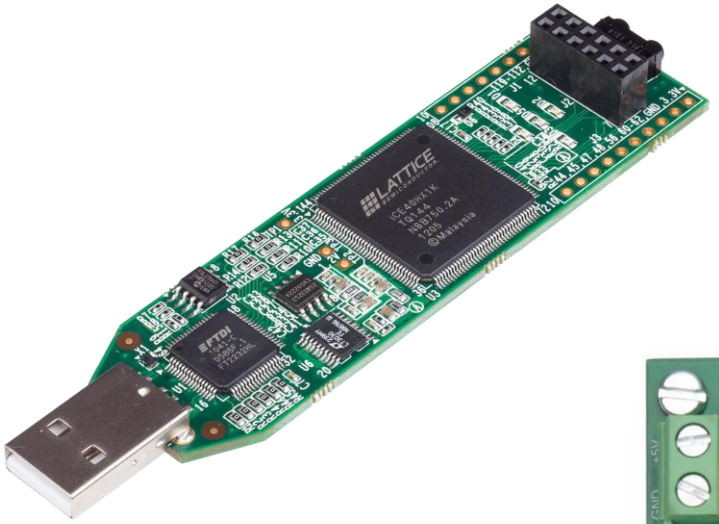


Raise your hand if:

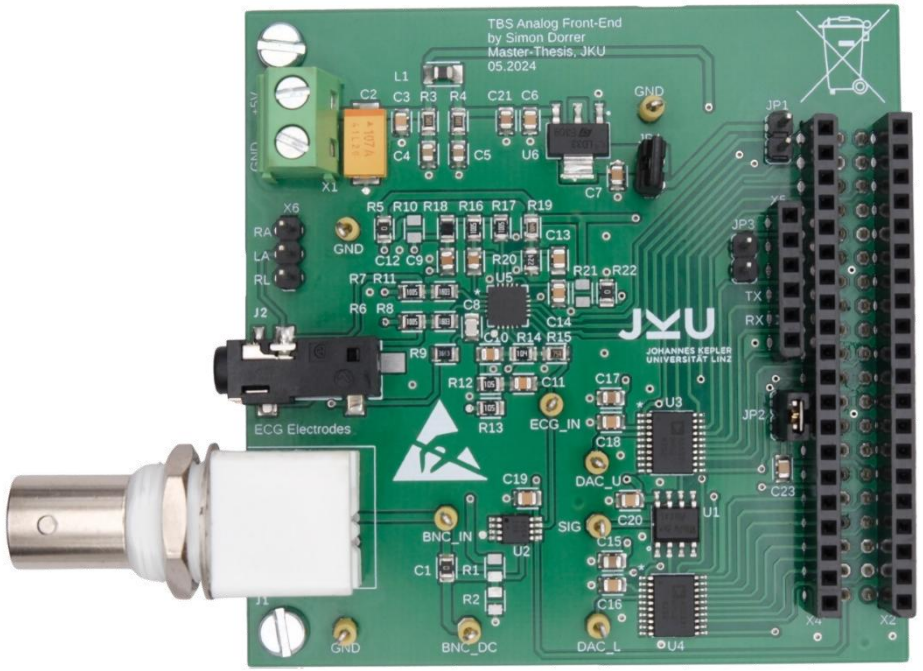
- You know what **VLSI** stands for
- You know what **ASIC** stands for
- You know what **HDL** stands for
- You have already used any HDL before
- You have already worked with **MCU**s or **FPGA**s
- You have already designed an **ASIC**
- You want to learn how to build a **tapeout**-ready chip
- You regularly use **AI agents**

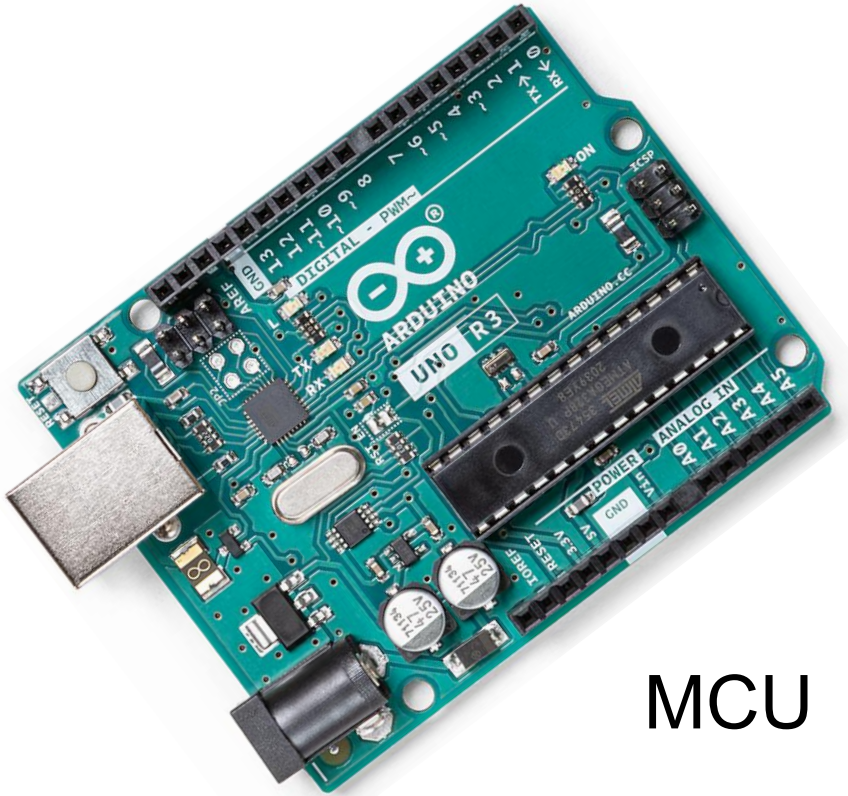


What are electronic systems?



FPGA



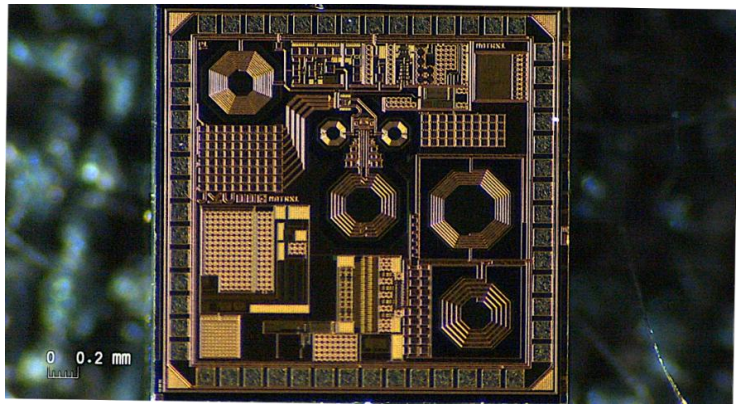


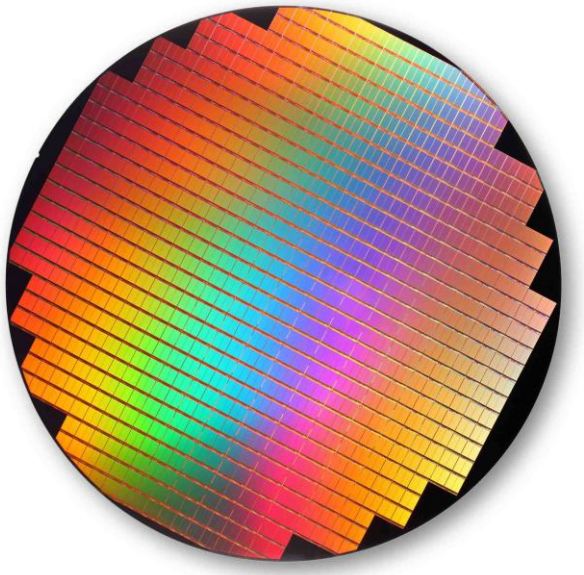
MCU

Mixed-Signal Designs

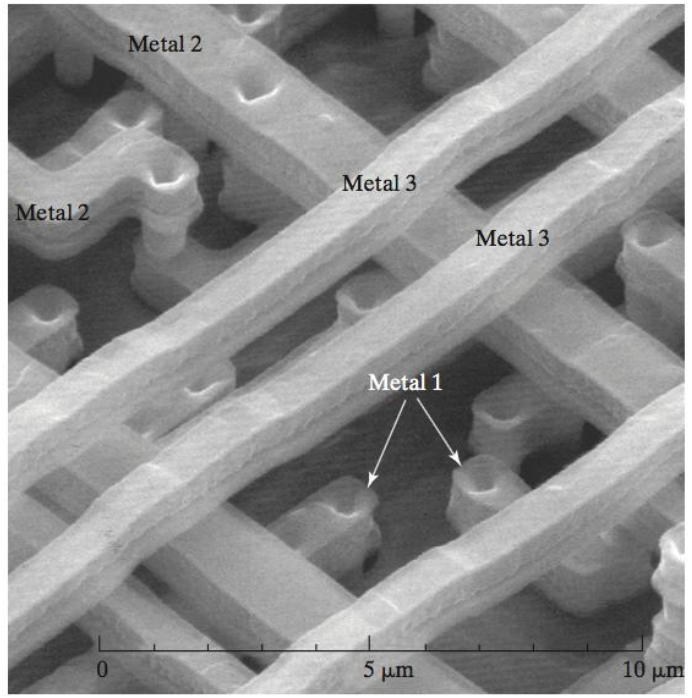
What are integrated circuits?

- Electrical circuits that are processed on silicon wafers.
- Components and interconnections are built up layer by layer through various processing steps.









https://www.allaboutcircuits.com/news/diamonds-vie-to-replace-silicon-as-next-semiconductor-material/

From Sand to Silicon: The Making of a Microchip



Overview & Goals



Requirements

- Bachelor CS / Master ELIT course (elective subject)
- Basic understanding of CMOS transistors
- Good understanding of digital logic (gates, flip-flops, etc.)
- Basic experience with a hardware description language (e.g., VHDL)
- Own laptop and ≥ 20 GB of free disk space (VSCode, Git, IIC-OSIC-TOOLS)
- Optional:
  - KV Analog Circuit Design: https://github.com/iic-jku/analog-circuit-design
  - KV Design of Complex Integrated Circuits: https://github.com/iic-jku/design-complex-ic

Content



- KV VLSI Design: https://github.com/iic-jku/vlsi-design
- Introduction
- From VHDL to SystemVerilog & modeling guidelines
- Functional verification with cocotb
- Logic synthesis, standard-cell library & equivalence checking
- Sequential elements & static timing analysis (STA)
- Synthesis optimizations & low-power design
- Clock domain crossing (CDC) & design for test (DFT)
- RTL-to-GDS flow with LibreLane, chip assembly & padframe
- Project: tapeout-ready RISC-V SoC

Grading

- **Part 1 – Theory (50%):**
  - Theoretical lectures mixed with small examples
  - Written exam before Christmas (**17.12.**)
- **Part 2 – Project (50%):**
  - Implementation of a full tapeout-ready RISC-V SoC

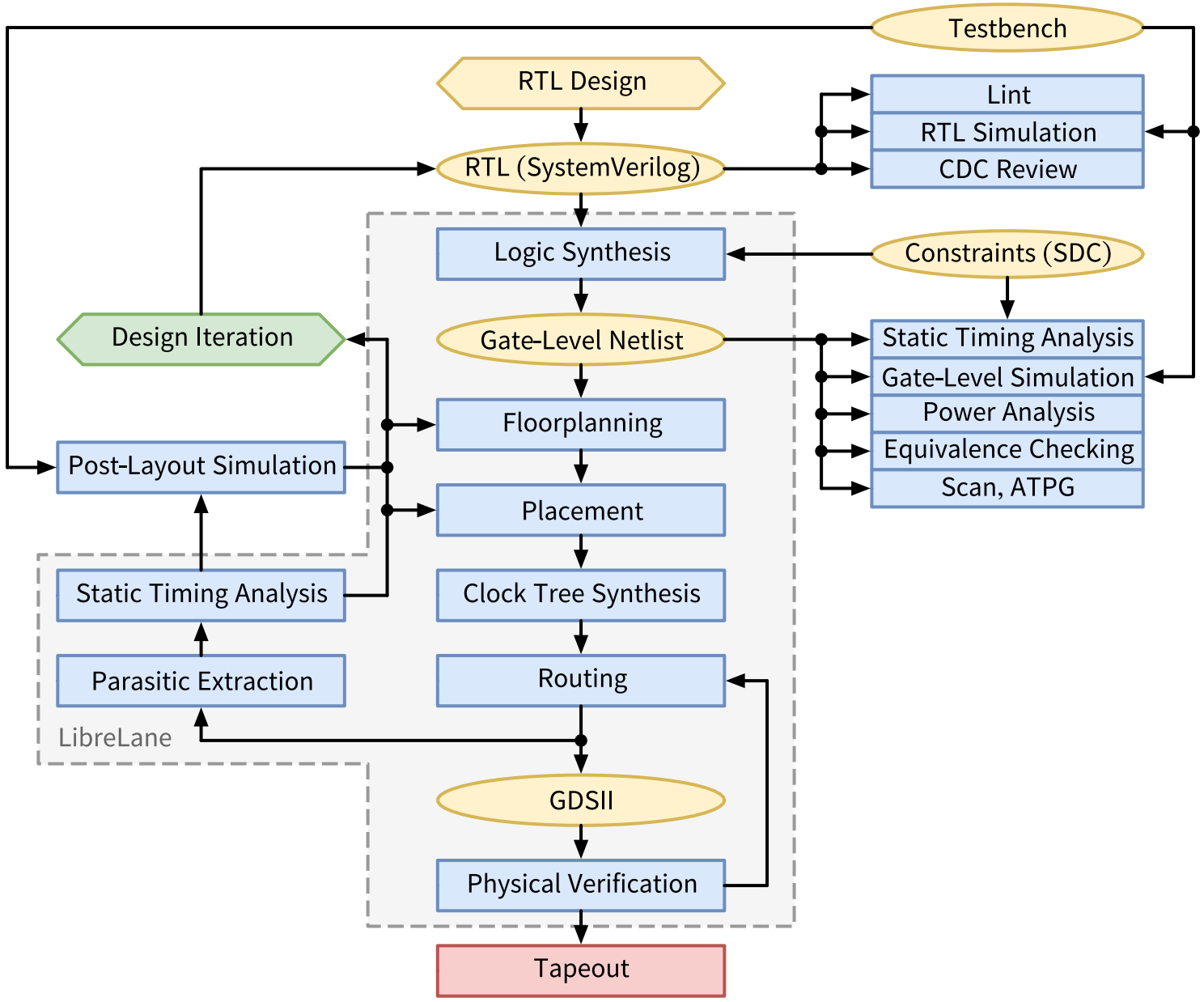
    + own extension (analog or digital)
  - Oral presentation
  - **The best project might be taped out!**
  - **Possible extension to seminar work (+3ECTS) or thesis!**

Goals? After this course you should:

- Be able to handle the IIC-OSIC-TOOLS
- Be able to handle GitHub
- Be able to read and write SystemVerilog RTL code
- Be able to verify RTL with self-checking cocotb testbenches
- Be able to understand and explain RTL-to-GDS flows
- Be able to handle timing (STA), testability (DFT), and clock domains
- Be able to build tapeout-ready ASICs with open-source tools
- Be able to use LLM agents efficiently (optional)

RTL to GDS Flow

- **Ovals**: design data
- **Boxes**: steps
- **Dashed**: LibreLane



Adapted from: S. Dorrer, “An Open-Source Adaptive Event-Based ADC for Bio-Signal Acquisition in 130 nm CMOS”, Master’s Thesis, Johannes Kepler University Linz, 2025. https://epub.jku.at/urn/urn:nbn:at:at-ubl:1-90281

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“The aha moment of seeing your circuit come to life will stay long after the equations are forgotten.” – Peter Kinget

Analog Mixed-Signal Design Flow & Template



GitHub Template

- **Solderpad Hardware License v2.1**
- Use this template or `git clone …`
- `make all`
- **Tapeout-ready AMS chip!**
- **Tutorial** based on Quarto
- **Regression tests** via GitHub actions
- **Unbelievable?**
  - **Try it out now!** →
- Tapeout proven?

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An Open-Source Analog Mixed-Signal Chip Design Template & Tutorial for the ihp-sg13cmos5l Open-PDK

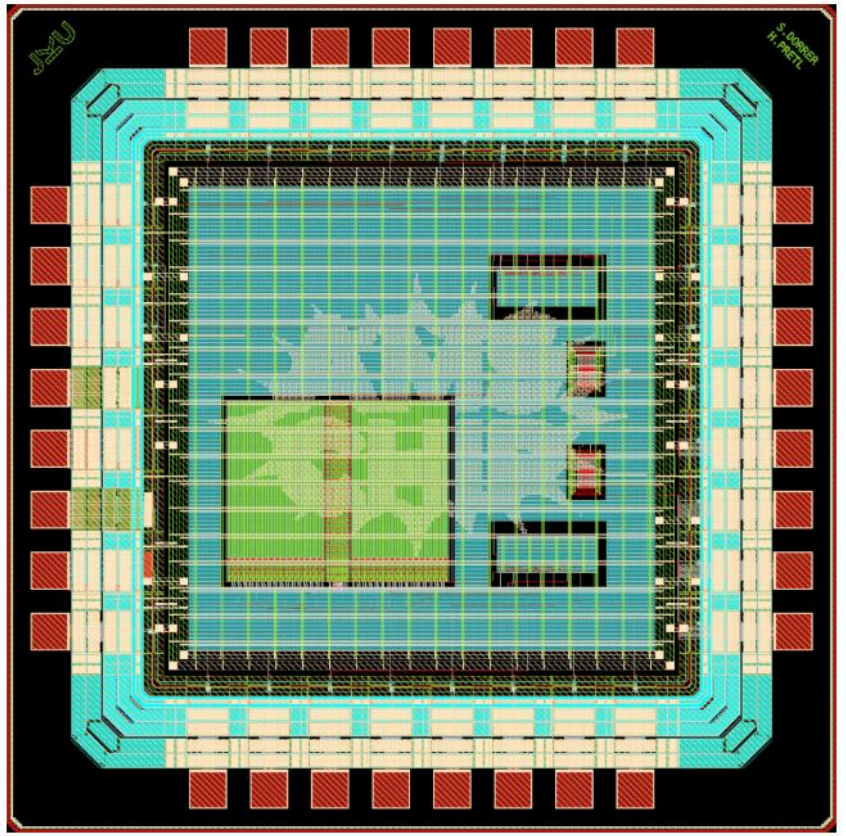
License Solderpad Hardware License v2.1 | quarto-publish passing | regression passing | License Check passing | Tutorial online

DOI 10.5281/zenodo.22115134

(c) 2026 Simon Dorrer and Harald Pretl

Institute for Integrated Circuits and Quantum Computing, Johannes Kepler University (JKU), Linz, Austria

> **Important**
>
> This repository requires the IIC-OSIC-TOOLS container with tag `2026.09` or later.



Chip render of the ihp-sg13cmos5l analog mixed-signal template chip (1.6 mm x 1.6 mm).

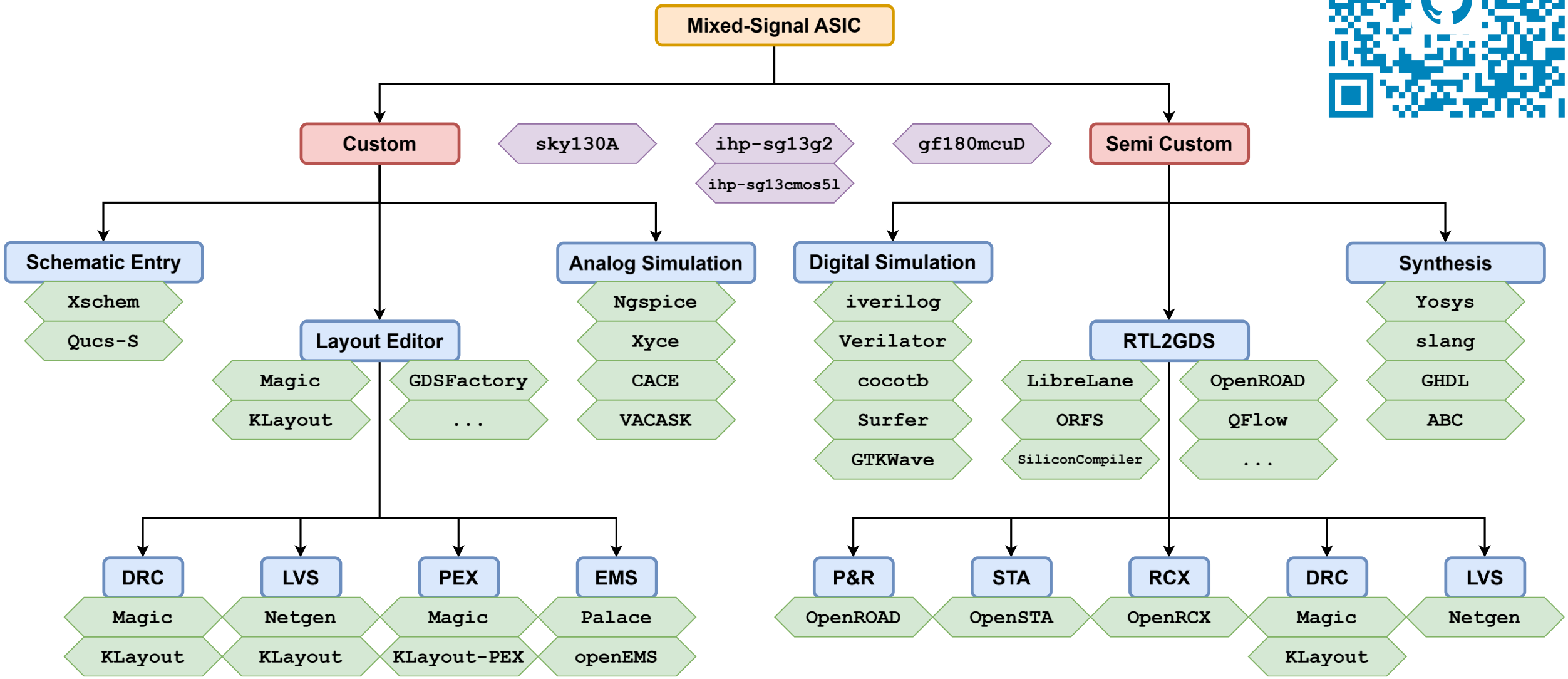
Flow Overview



- Simulation, linting, and hardening of RTL
- Layout checks and reports of hardened digital macro
- Simulation of all analog testbenches (incl. PVT and MC)
- Layout checks of analog & digital blocks (DRC, LVS, and PEX)
- Digital-on-top top-level assembly and layout check (DRC)
  - Seal ring and pad frame generation
  - Macro placement and wiring also with non-default rules (NDRs)
  - Logo and fill insertion
- Full top-level LVS and simulation without black boxing
- Bonding diagram

IIC-OSIC-TOOLS

IIC-OSIC-TOOLS
OPEN-SOURCE IC DESIGN · JKU LINZ



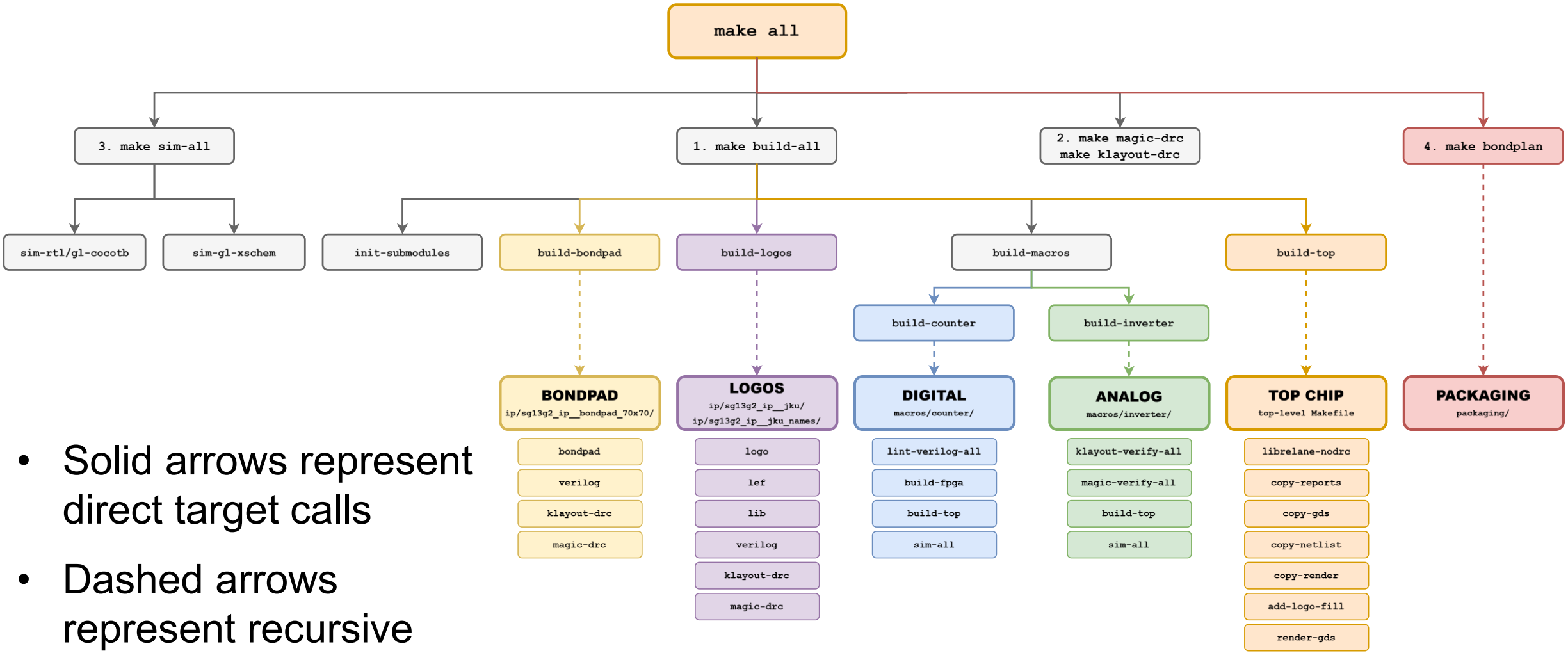
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Folder Structure

- Structure discussed with FOSSi community
- Easily share and reproduce results of other macros



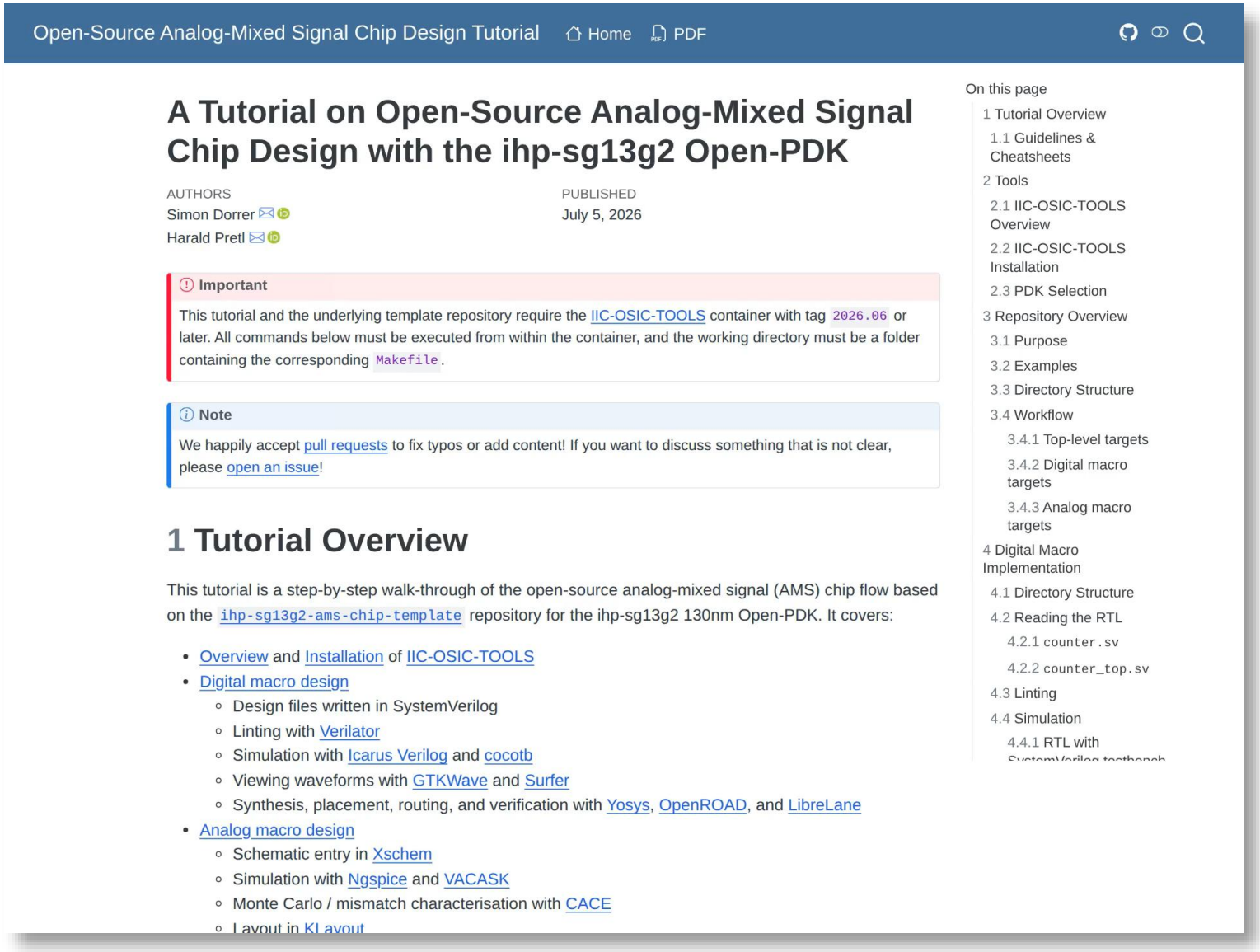
Makefile Structure



- Solid arrows represent direct target calls
- Dashed arrows represent recursive target calls

Tutorial

- Walk through all IC design steps to get started
- Using IHP's 130 nm CMOS process
- Simple exercises to learn by doing



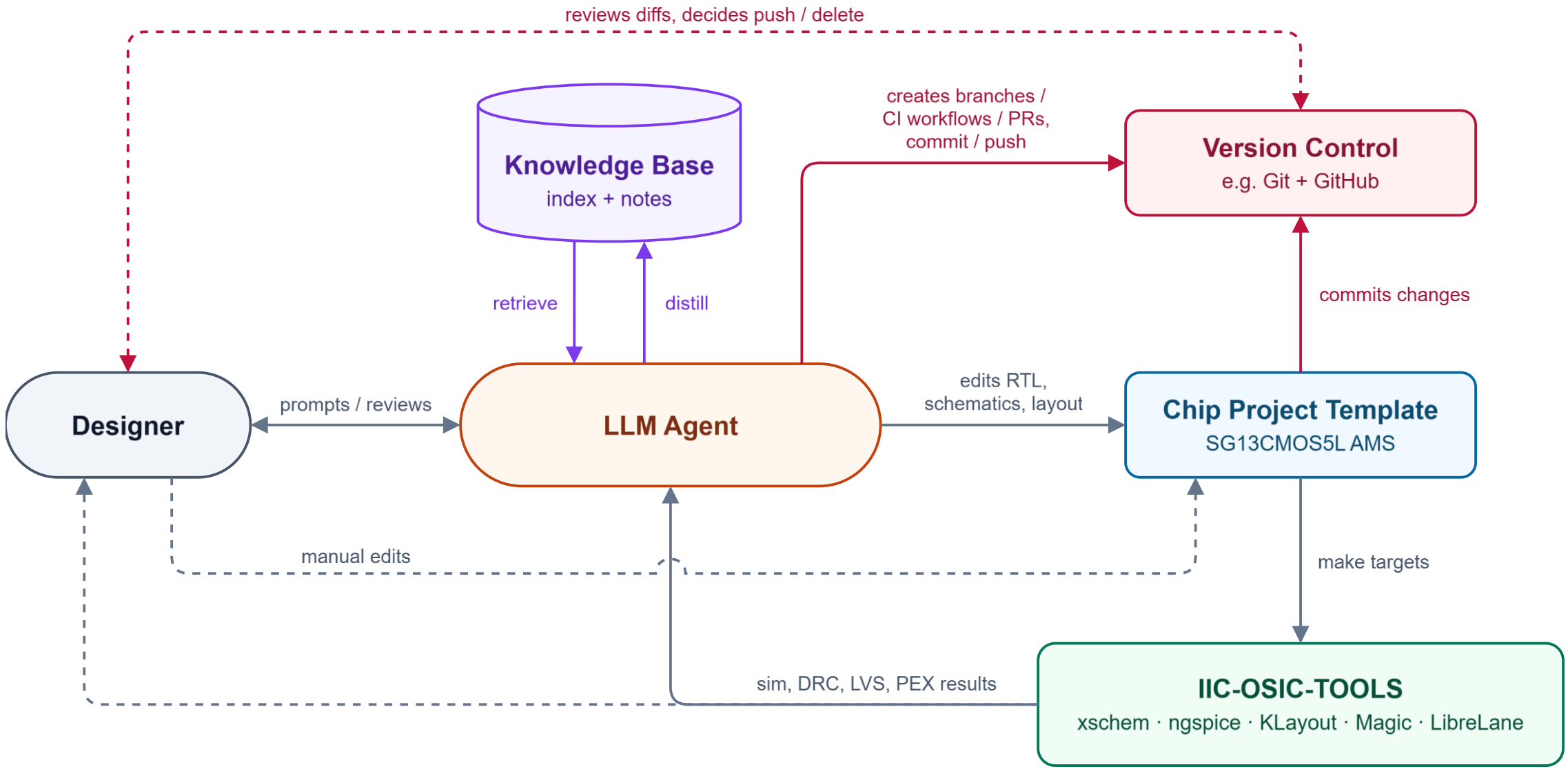
Agentic AI Chip Design



Are you allowed to use AI?

- **Yes,** as a **learning partner**, not as a substitute for your own thinking!
- **Written exam:** no aids, only what you have mastered counts
- **Oral presentation:** you have to explain what you did and why
- **First** try to understand and solve a problem on your own
- **Then** use AI to check your solution, see alternatives, or explain a topic
- **Goal:** the competence to get there yourself, not the fastest answer
- **In the end, you want to learn something. Do not cheat yourself!**
- **We always need people who understand how to prompt and review the results.**

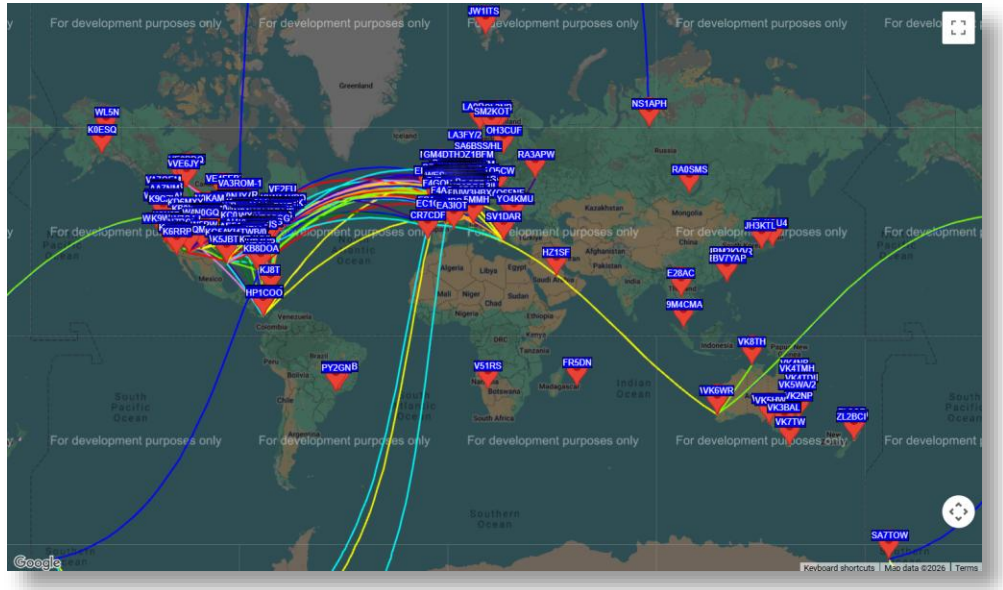
Workflow


reviews diffs, decides push / delete
Knowledge Base
index + notes
creates branches /
CI workflows / PRs,
commit / push
Version Control
e.g. Git + GitHub
retrieve
distill
commits changes
Designer
prompts / reviews
LLM Agent
edits RTL,
schematics, layout
Chip Project Template
SG13CMOS5L AMS
manual edits
make targets
sim, DRC, LVS, PEX results
IIC-OSIC-TOOLS
xschem · ngspice · KLayout · Magic · LibreLane

TinyWhisper SoC

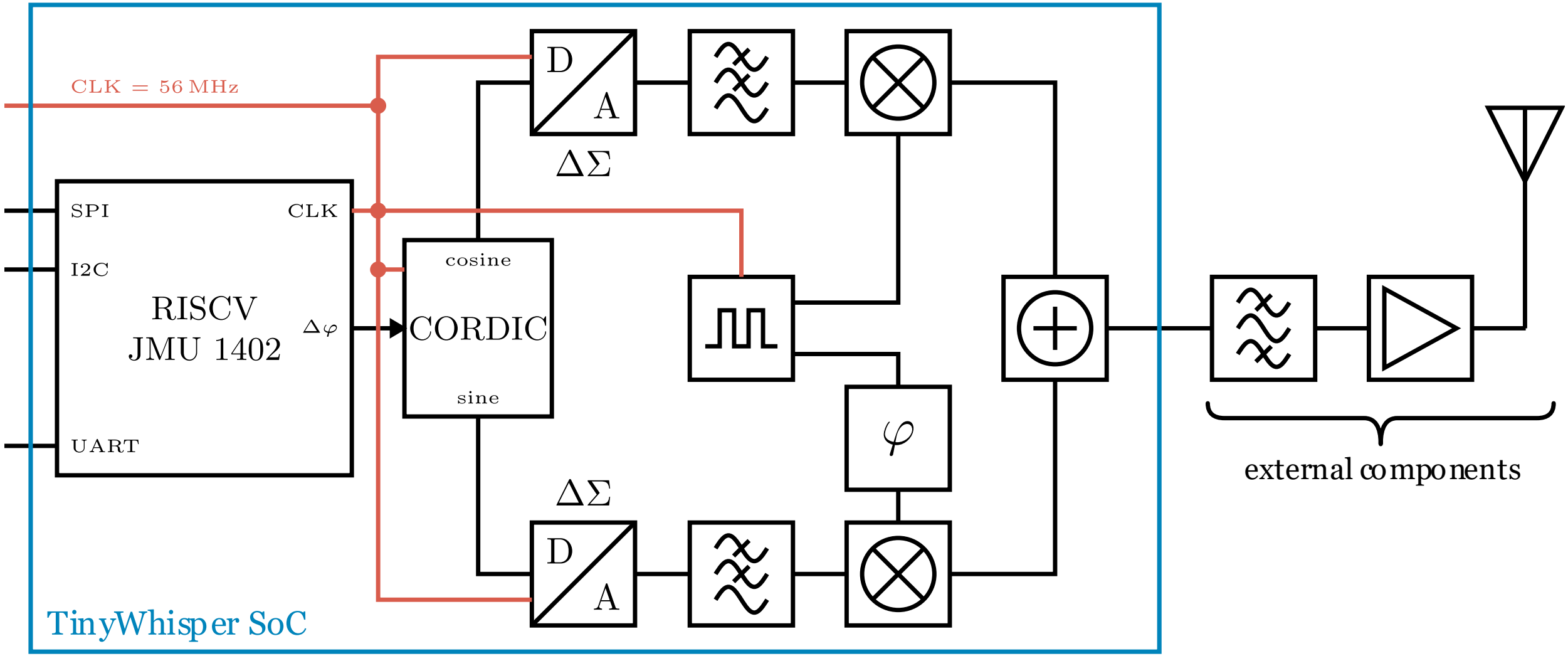


TinyWhisper SoC



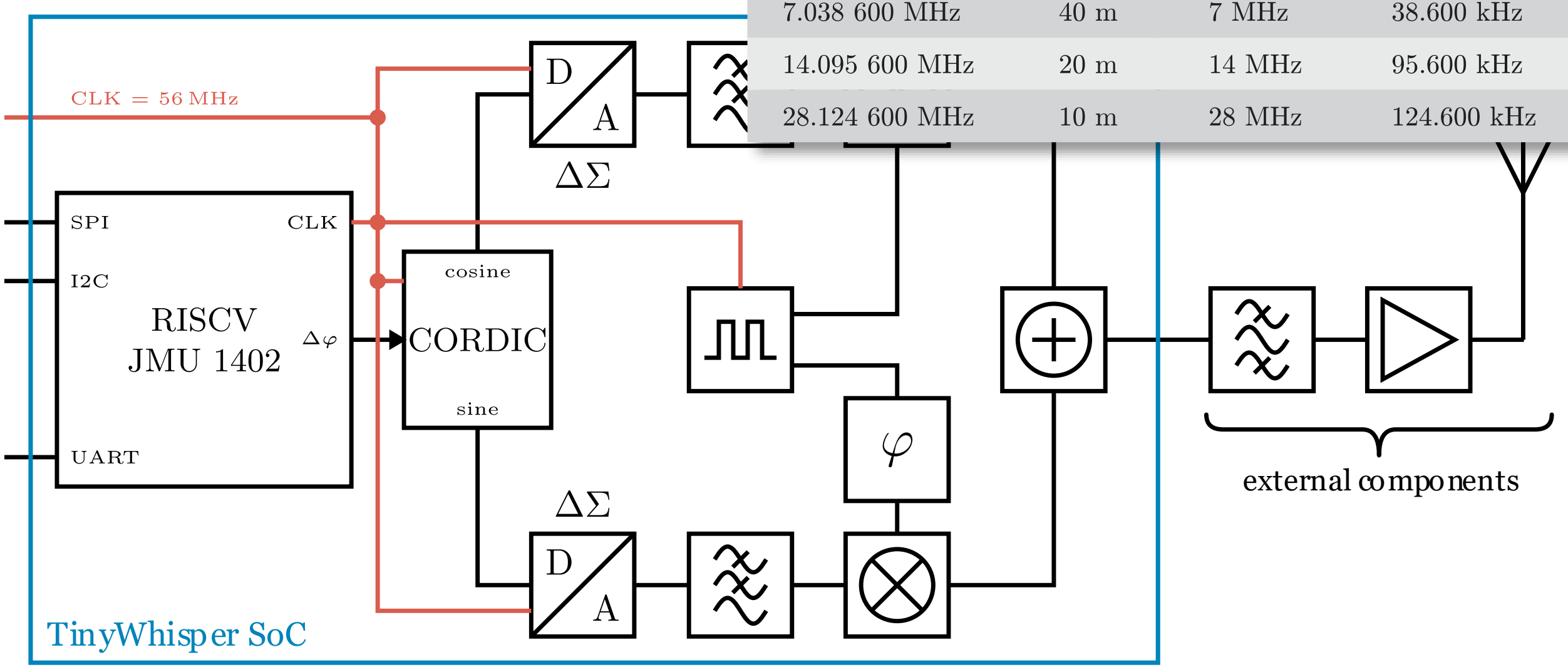
- Cooperation: JMU Würzburg & JKU Linz
- Weak Signal Propagation Reporter (WSPR):
  - Popular digital mode in ham radio
  - Propagation studies @ 80 m to 10 m bands
  - Transmits short message (callsign, location, and power)
  - Receivers report to central database via Internet
  - WSPRnet: https://www.wsprnet.org/drupal/wsprnet/map
  - 4-FSK modulation (Δf = 1.46 Hz, T = 0.683 s)
  - SNR down to -31 dB @ 2500 Hz
  - MODEM (coding, framing, symbol mapping) in FW (C) on RISC-V CPU

Block Diagram



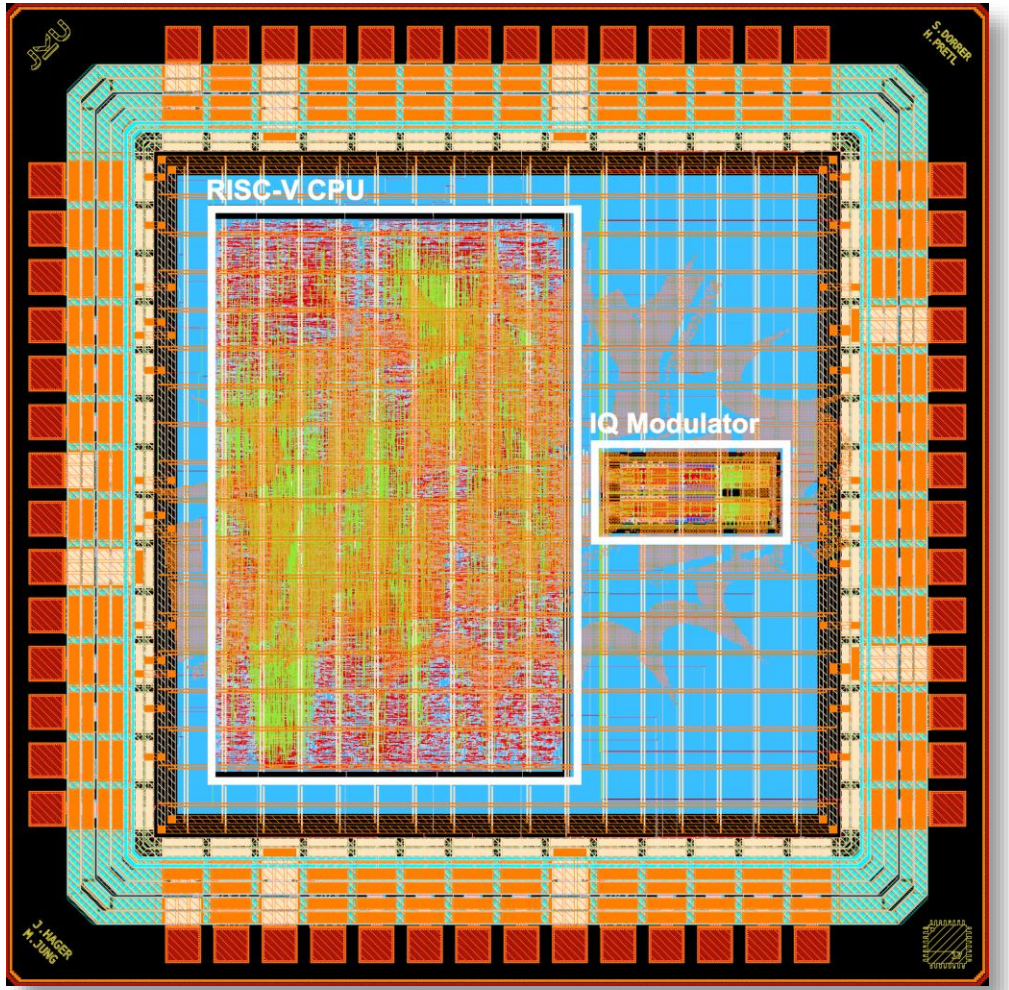
Block Diagram

RF	Band	LO	IF
3.592 600 MHz	80 m	3.5 MHz	92.600 kHz
7.038 600 MHz	40 m	7 MHz	38.600 kHz
14.095 600 MHz	20 m	14 MHz	95.600 kHz
28.124 600 MHz	10 m	28 MHz	124.600 kHz



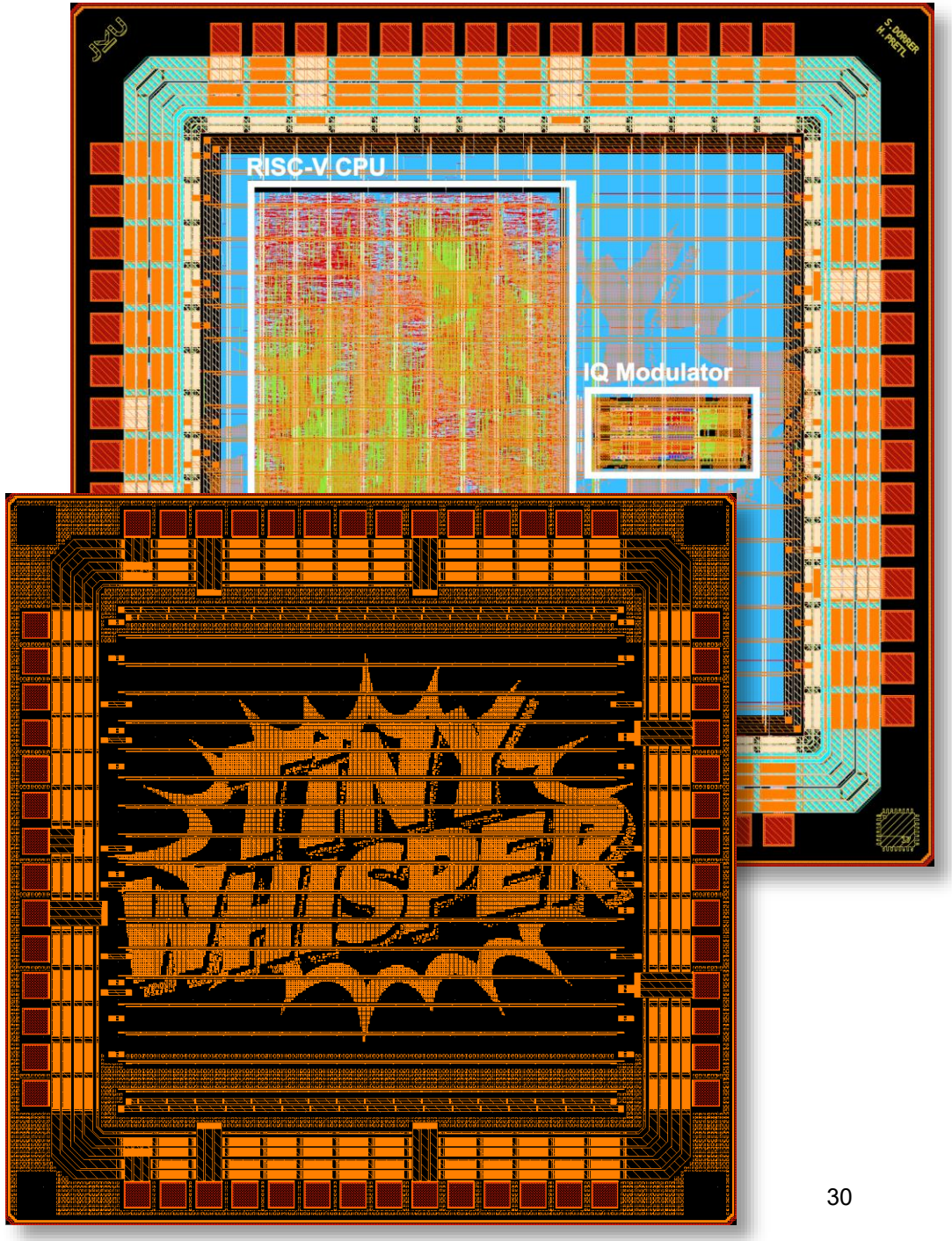
Overview

- 130 nm IHP SG13CMOS Open-PDK
- 2000 µm × 2000 µm (4 mm²)
- 1270 µm × 1270 µm (1.61 mm²)



Overview

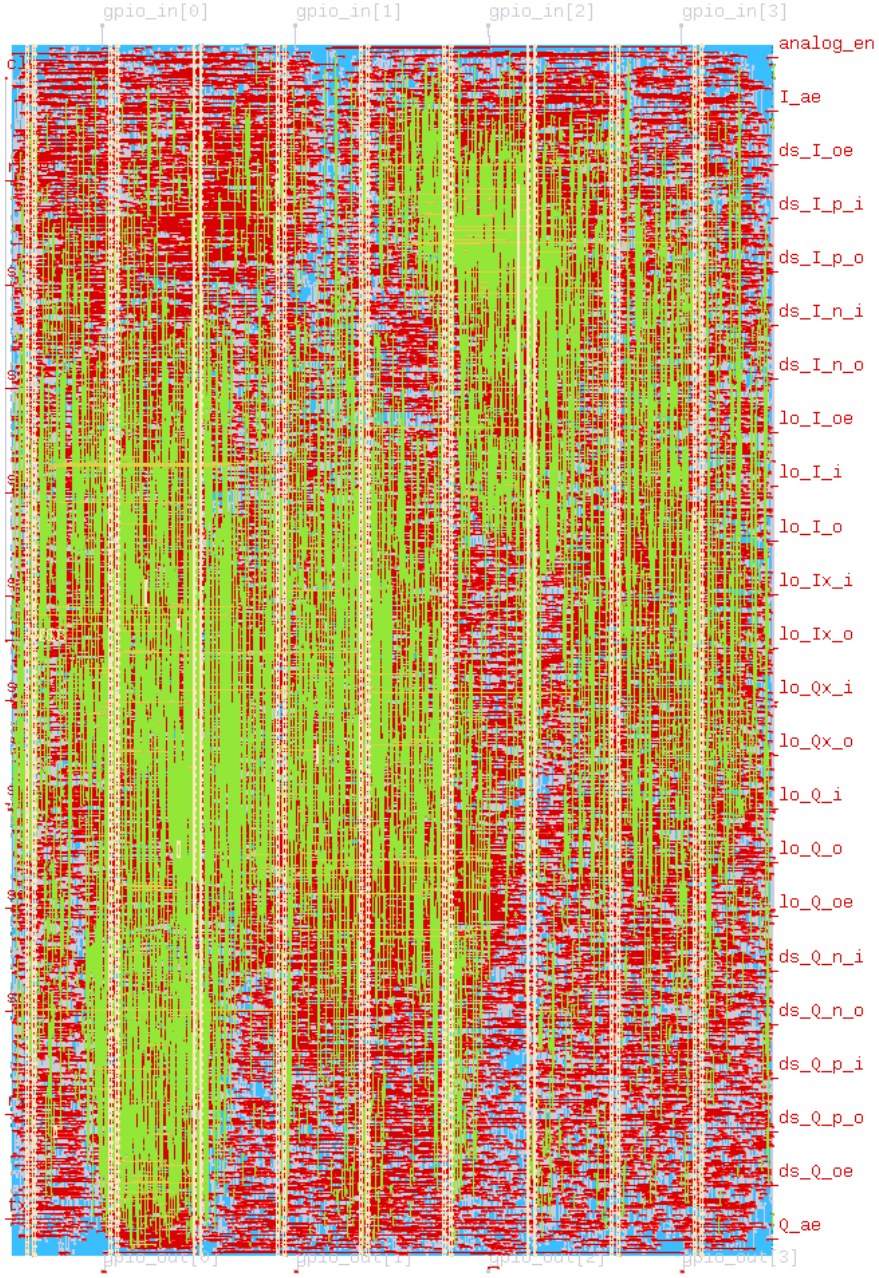
- 130 nm IHP SG13CMOS Open-PDK
- 2000 µm × 2000 µm (4 mm²)
- 1270 µm × 1270 µm (1.61 mm²)
- Logo with ArtistIC by PULP platform
- QFN-48 package
- 56 pads (8 bonded to package ground)
- 1.5 V supply voltage
- 3.3 V I/O supply voltage
- **Chip in production!**



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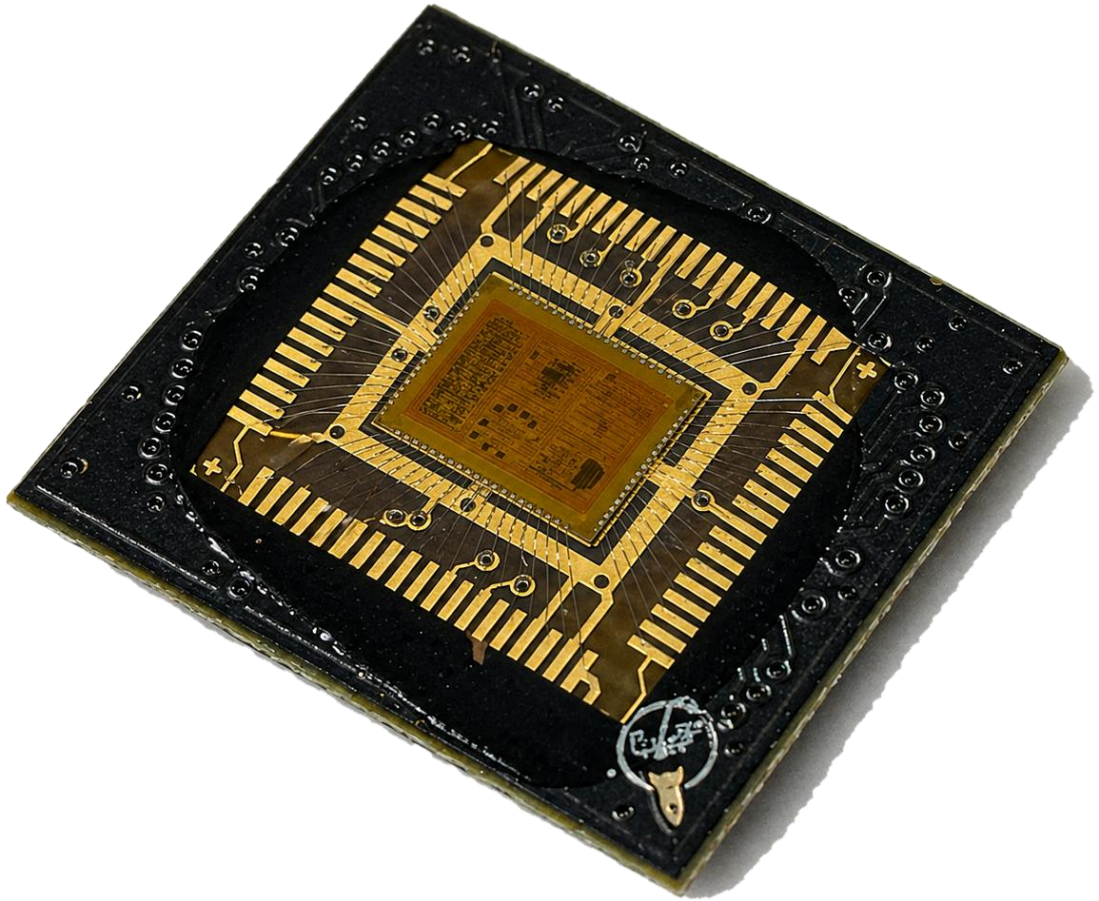
RISC-V CPU

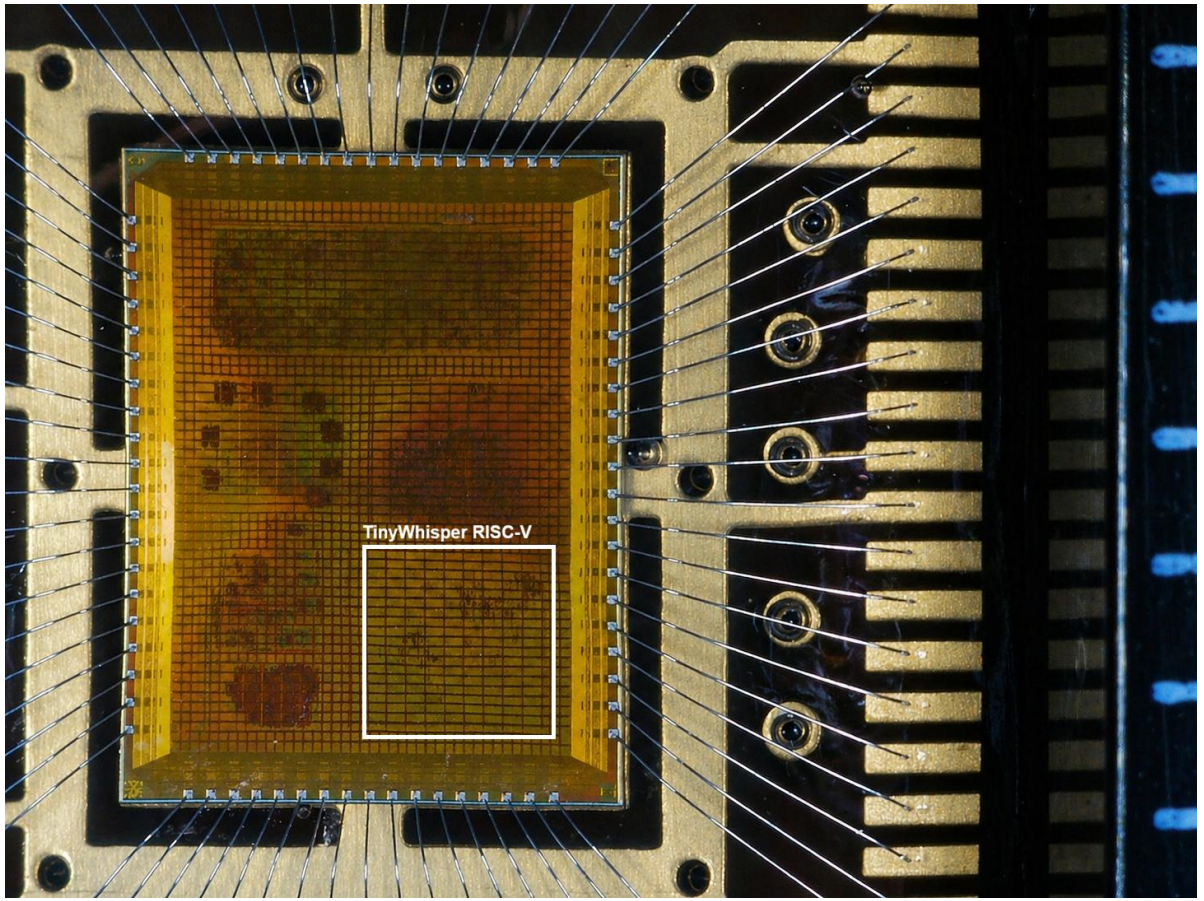
- RISC-V *rv32e_zicsr* embedded CPU
  - Multi-cycle, no pipelining
  - 56 MHz clock frequency
  - SPI, I2C, UART, GPIO are memory-mapped I/O
- Baseband signal generation:
  - 30-bit iterative CORDIC ($\Delta f = 0.05$ Hz)
  - Single-bit DSM (2nd order, OSR ≤ 256)
  - Configurable via software
- 25 % duty cycle LO generation for mixer



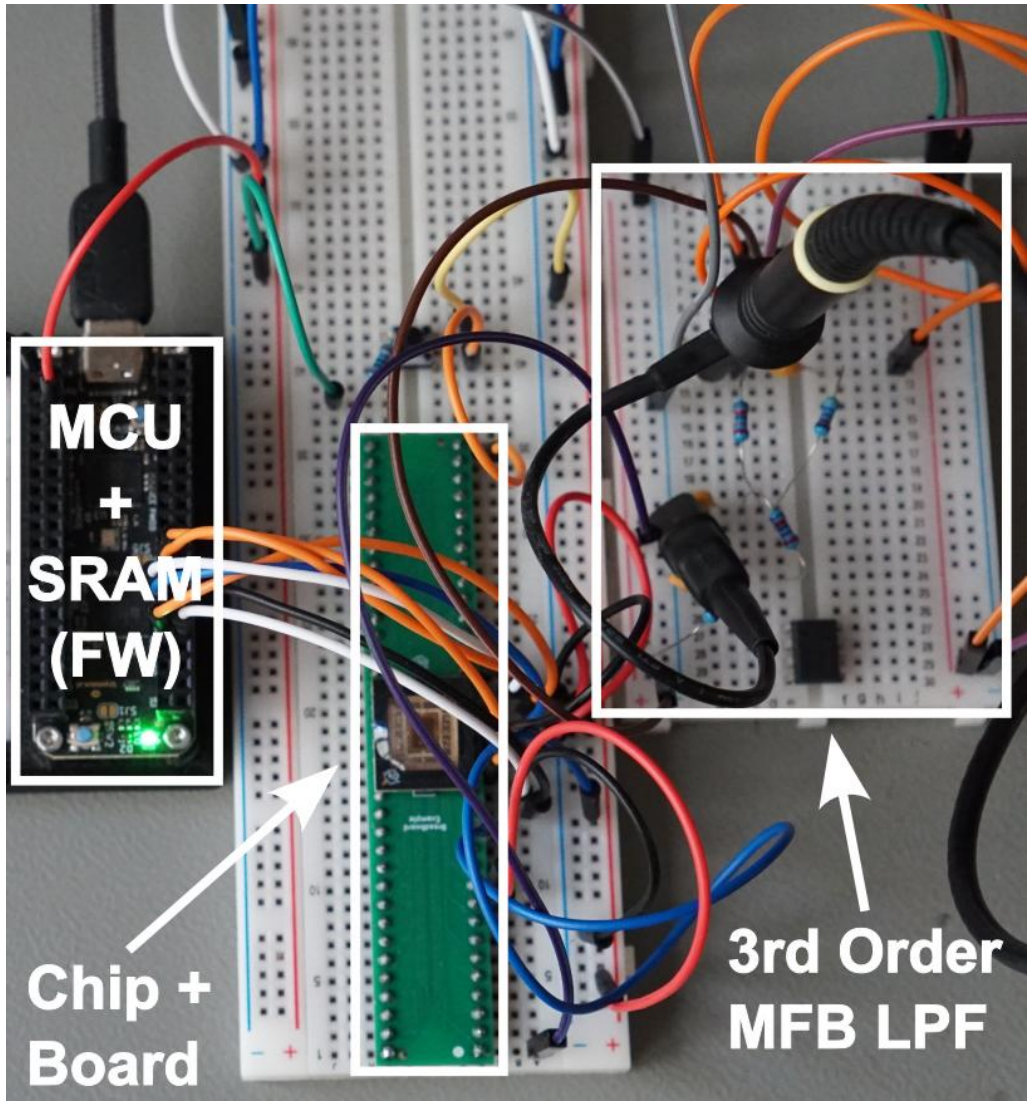
RISC-V CPU – wafer.space Run 1

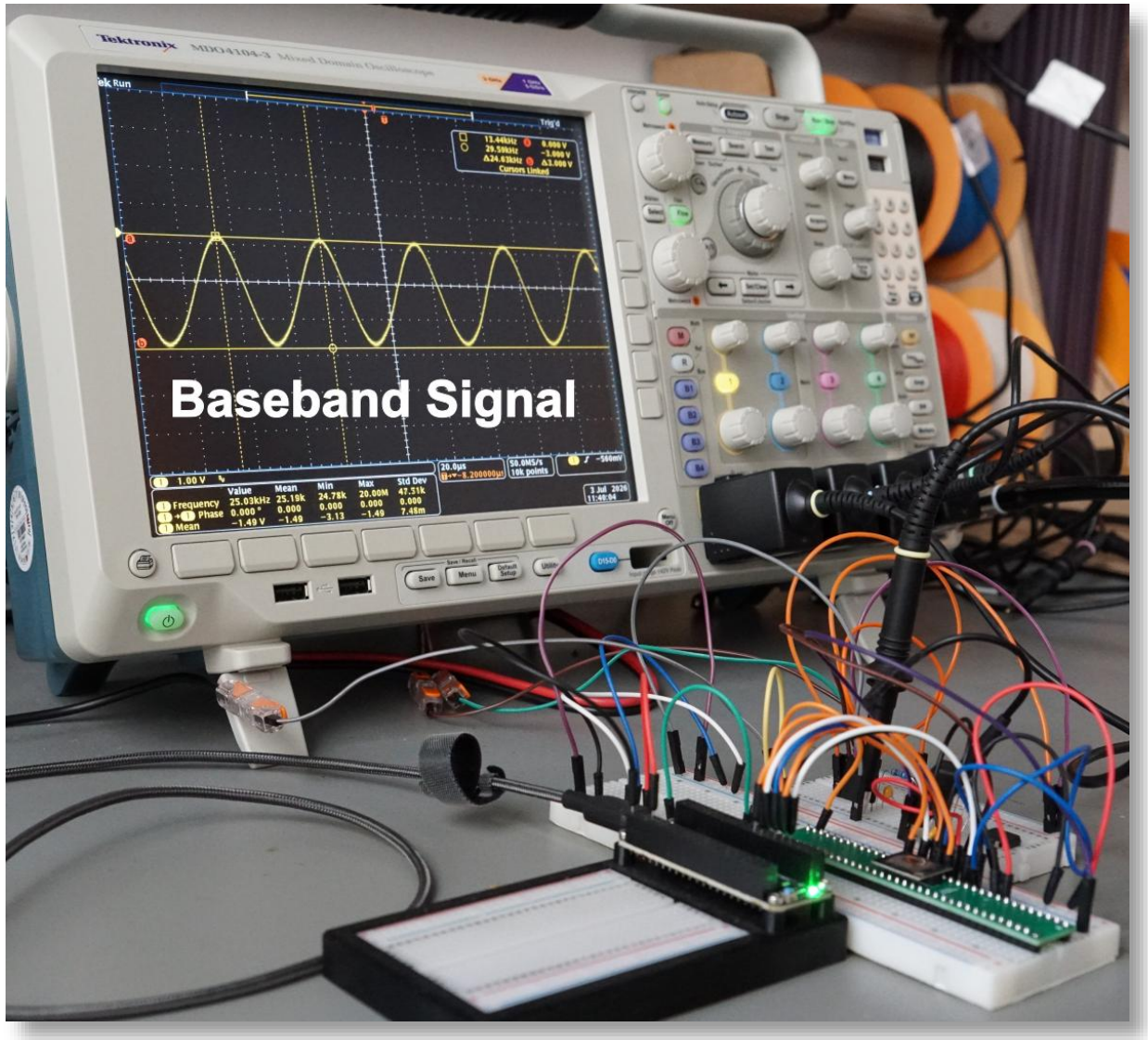
- GlobalFoundries GF180MCU
- Multi-project student tapeout (digital filters, mini games, …)





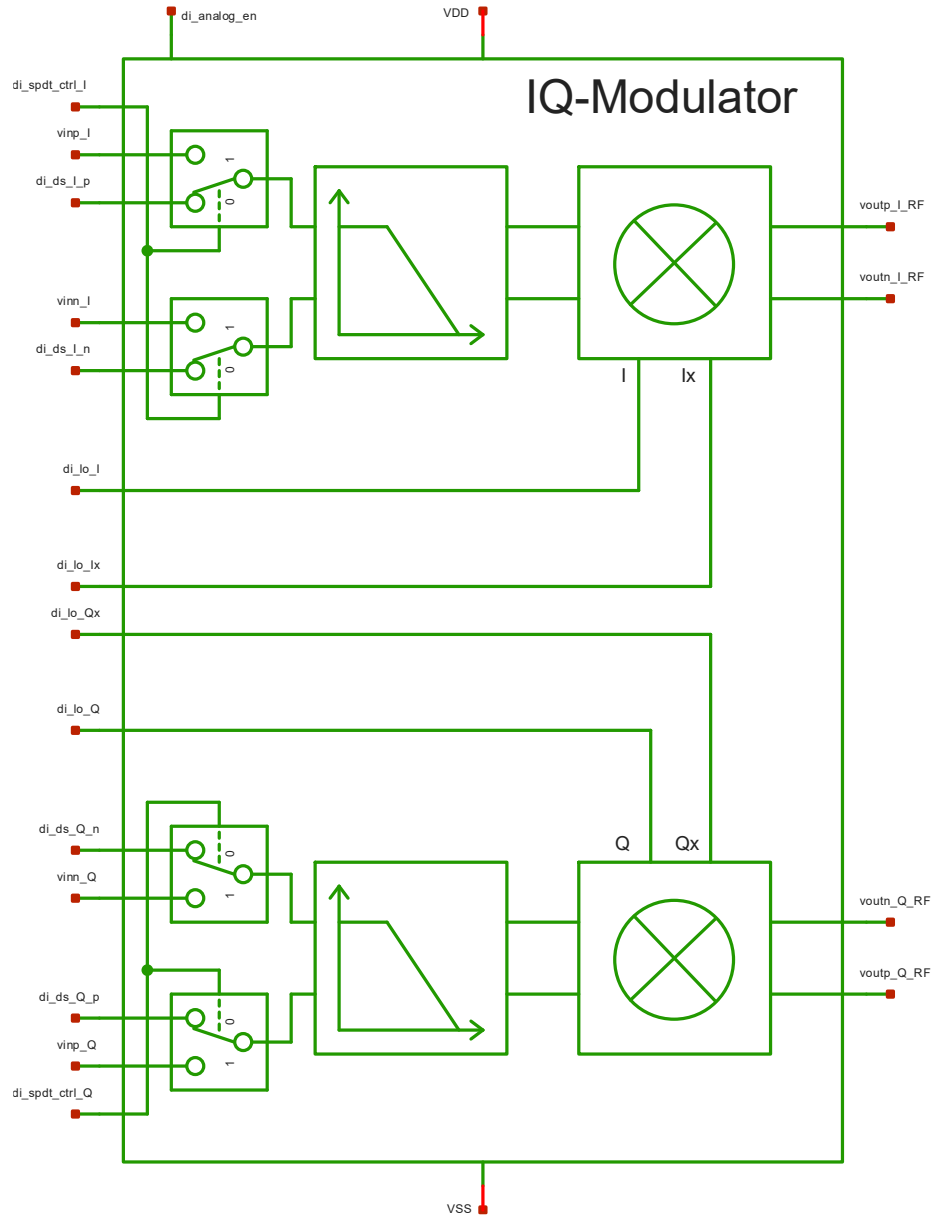
RISC-V CPU – wafer.space Run 1



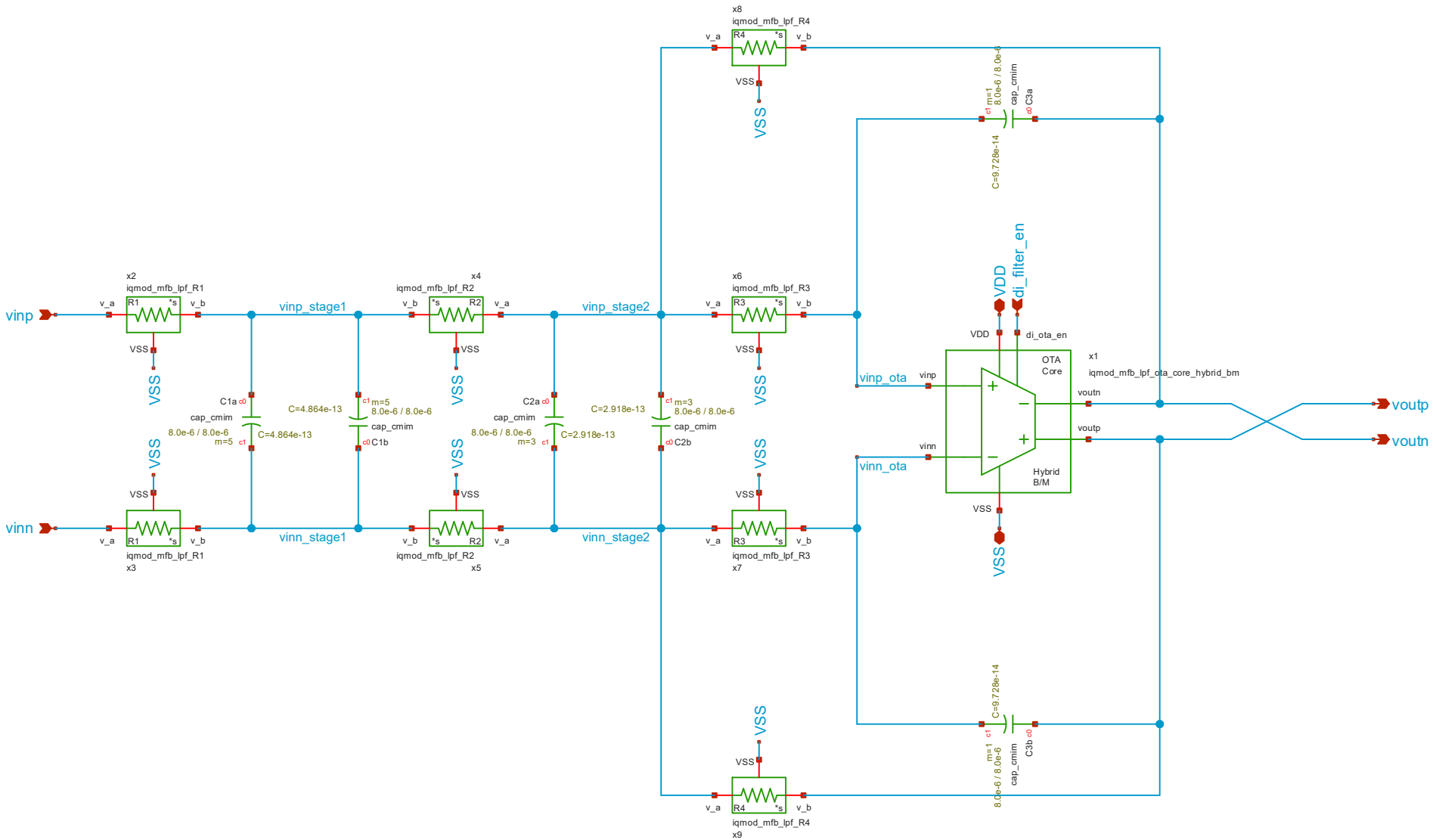


IQ Modulator

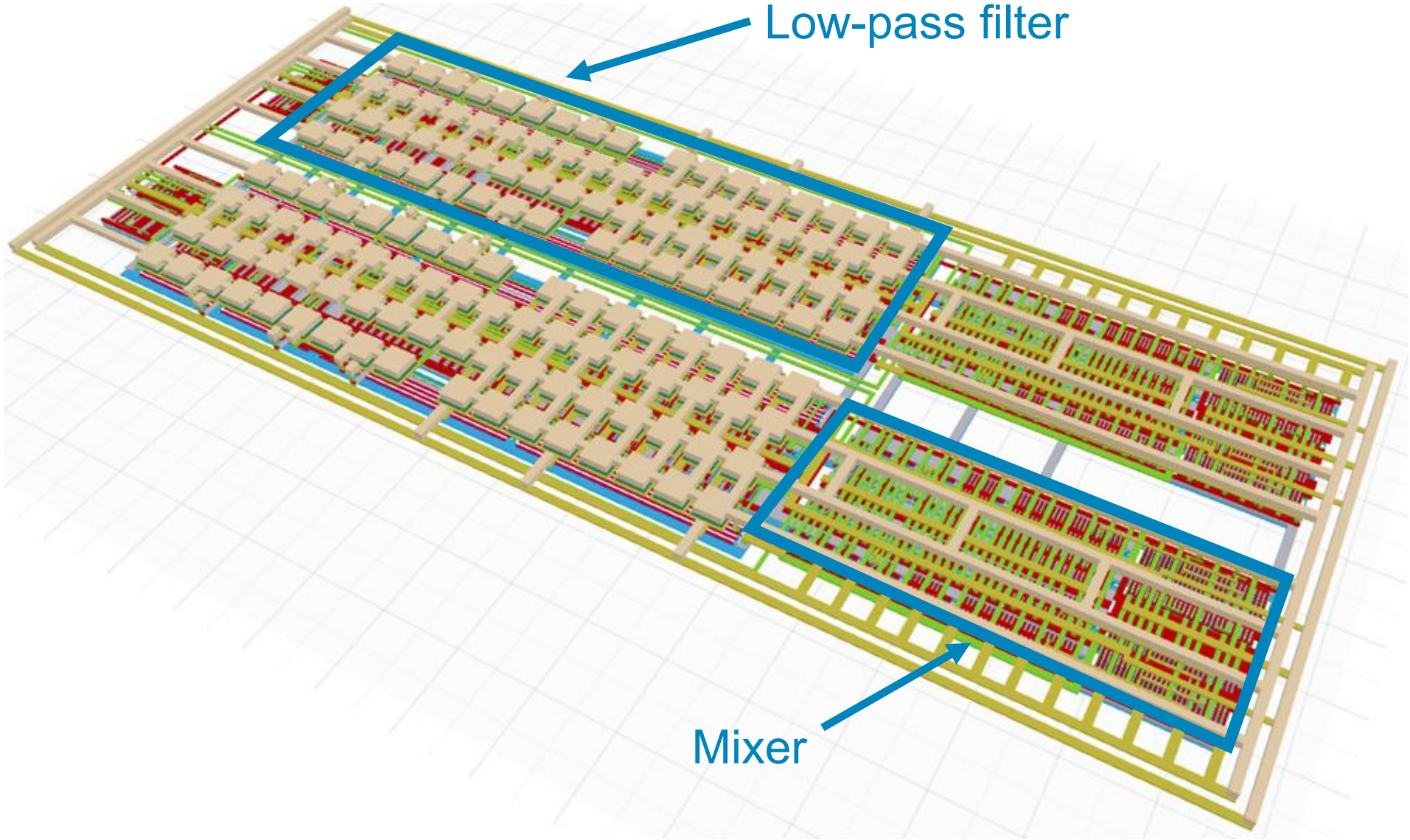
- 3rd order multiple-feedback low-pass filter
  - Butterworth
  - 400 kHz cutoff frequency
  - Barthélemy / Manfredini inverter-based OTA
- Passive CMOS voltage-mode mixer
  - Simple transmission-gate (TG) switch
  - Single-to-differential conversion circuit
- Only inverter- and TG-based circuits
- IQ-TX: Not only WSPR, but FT8, SSB, CW, …



Schematic Drawing is Art!

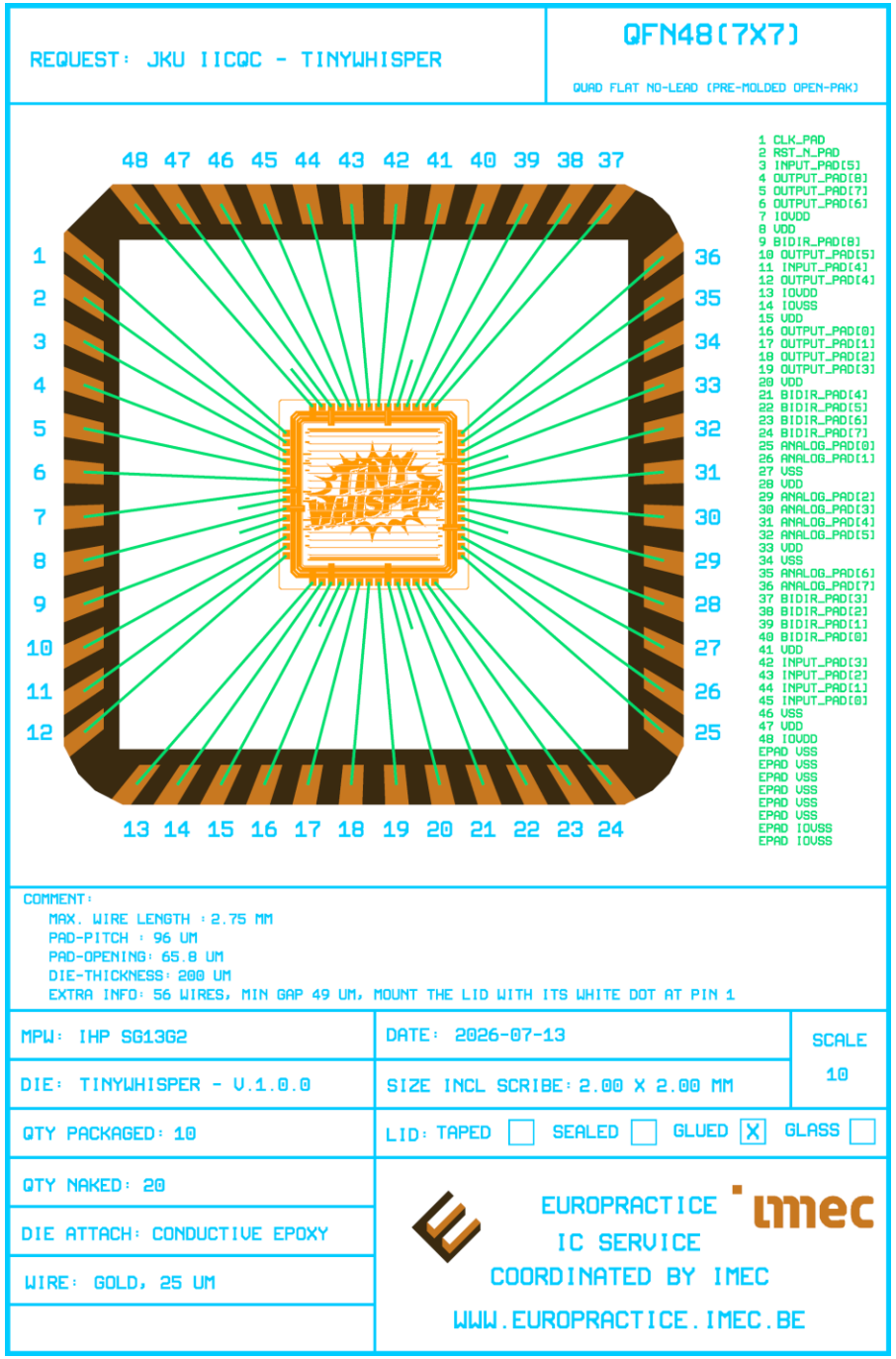


IQ Modulator



Bonding Diagram

- Automatic generation:
  - Python script
  - Config file (pinout, placement)
  - Extract layers (TM2, pads)
  - Place chip layout
  - Get pad centers
  - Draw bond wires
  - Render GDS



Tools Installation



Local Installation

1. Install **VS Code**: https://code.visualstudio.com
2. Run the **installer** (Git, Docker, WSL2 / XQuartz, IIC-OSIC-TOOLS):
   - Windows (admin PowerShell): `irm https://osic.tools/install.ps1 | iex`
   - macOS / Linux: `curl -fsSL https://osic.tools/install.sh | bash`
3. Reboot, then start Docker Desktop once
4. Start the container in the `iic-osic-tools` folder:
   - Windows: `.\start_vnc.bat` macOS / Linux: `./start_vnc.sh (or start_x)`
   - Open http://localhost/?password=abc123 (first start pulls ~4 GB)
5. Your designs: `~/eda/designs` (laptop) = `/foss/designs` (container)

Containerized On-Demand Workspace System (COWS)

1. Open COWS: https://jku128011.riic.uni-linz.ac.at/
2. Log in and change password
3. Create a workspace
    a. Go to “Workspaces” and click “Create workspace”
    b. Enter “Workspace name” and select “KV VLSI Design 26/27” template
4. Start the workspace → “Status” should change to “Running”
5. Open it by clicking the workspace name
6. **Enjoy your workspace! Happy designing!**



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